In-Class Practice: Oral Absorption

Module 5 - Single Oral Administration · In-class activity · 21 Sep 2026

Name _____ Date _____

Problem 1 A drug given orally has a half-life of absorption of 90 minutes. What is its absorption rate constant?

Problem 1 A 500-mg dose of a drug that is 88% bioavailable is given by mouth. Its half-life of absorption is 90 minutes and its half-life of elimination is 5 hours, and its apparent volume of distribution is 20 L. When does the peak concentration occur?

Problem 1 A 500-mg dose of a drug that is 88% bioavailable is given by mouth. Its half-life of absorption is 90 minutes, its half-life of elimination is 5 hours and its apparent volume of distribution is 20 L. What is the peak plasma concentration?

Problem 1 For that same 500-mg oral dose, if the apparent volume of distribution were 40 L rather than 20 L, what peak plasma concentration would be expected?

Problem 2 A drug given as a 750-mg oral dose gives plasma concentrations described by $C_p = 23.2(e^{-0.182t} - e^{-0.872t})$, with t in hours. What is its elimination half-life?

Problem 2 A drug given as a 750-mg oral dose gives plasma concentrations described by $C_p = 23.2(e^{-0.182t} - e^{-0.872t})$, with t in hours. Calculate t_{max}.

Problem 2 A drug given as a 750-mg oral dose gives plasma concentrations described by $C_p = 23.2(e^{-0.182t} - e^{-0.872t})$, with C_p in mg/L and t in hours. Calculate C_{max}.

Problem 2 A 750-mg oral dose of a drug that is 84% bioavailable gives plasma concentrations described by $C_p = 23.2(e^{-0.182t} - e^{-0.872t})$, with concentration in mg/L and time in hours. What is the apparent volume of distribution?